



Sunspot Number V2.0 Through Solar Cycle 25

A Long-term Multi-Proxy Stability Analysis

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The Sunspot Number is the longest direct index of solar activity.

- SN V2.0 is the modern reference scale released after major recalibration in 2015.
- SN is network-constructed index: Stability depends on counting rules (observer), coverage, instrumentation, and network evolution.

Context: Clette & Lefèvre (2016); Clette et al. (2023); SILSO Sunspot Number V2.0.

Does SN V2.0 show coherent drift relative to independent solar-activity proxies?

- Independent proxies provide external checks on whether SN behaves consistently.
- We have inspected 5 proxies that sample different layers of the Sun.
- Strong evidence requires sustained, multi-proxy residual structure.

Note

Coherent multi-proxy behavior matters more than any single-proxy excursion.

Steps for Detecting Drift

1 Data

Align monthly data

Common timestamps, monthly means, smoothing (centered 10 months running mean).

2 Diagnostic

Measure the fractional proxy residual

Fit proxy from SN

Each proxy gets its own SN-proxy relation.

Reconstruct proxy

Ask what the proxy should look like if SN explains it.

3 Detection

Inspect coherence

Look for persistent departures shared by several proxies.

Steps for Detecting Drift: Data

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Dataset: Reference and Photospheric Records

Series	Span used	Meaning	Role here	Source
SN V2.0	1818–present	Observer/network sunspot index	Reference scale under test	SILSO; Clette & Lefèvre
Sunspot area	1874–present	Photospheric area coverage	Long geometric proxy	Mandal et al. 2020
LOS magnetic field	1996–present	Photospheric magnetic signal	Magnetic-field context; shorter span	Scherrer 1995; Schou 2012

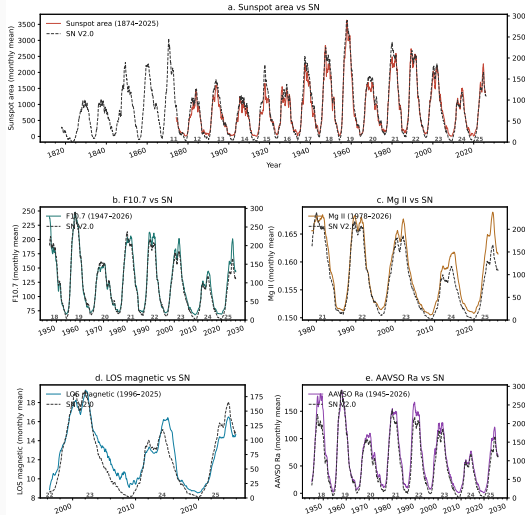
The photospheric records are closest to the spot-counting problem, but they still measure different quantities: counts, area, and magnetic flux.

Dataset: Radiative and Observer-Chain Checks

Series	Span used	Meaning	Role here	Source
F10.7	1947–present	10.7 cm radio flux	Upper-atmosphere activity proxy	Tapping 2013; Clette 2021
Mg II	1978–present	UV core-to-wing index	Chromospheric proxy	Viereck; de Toma; Bremen
AAVSO R_a	1945–present	Independent sunspot counts	Observer-chain comparison	AAVSO: Shapley–Hossfield

These records should follow the solar cycle, but their response to sunspots need not be linear or identical across cycle phase.

Data: All Proxies Follow the Solar Cycle



- Different start dates define different overlap intervals.

Steps for Detecting Drift: Diagnostic

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Reconstruct proxy

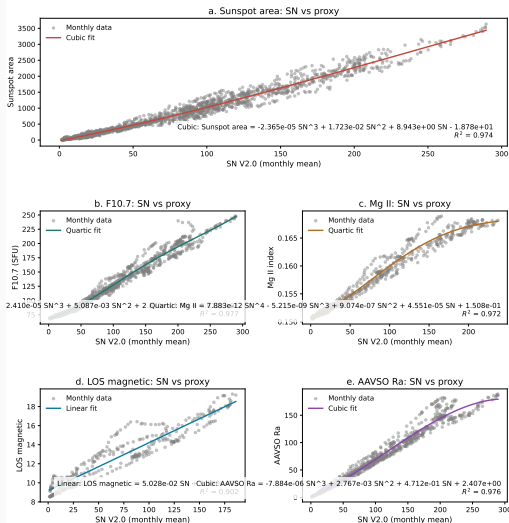
Ask what the proxy should
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Method: Nonlinear Relations Matter



- Proxy response to SN is not always linear.
- F10.7 is especially sensitive to averaging and low-activity curvature.
- Selected families: area cubic, F10.7 quartic, Mg II quartic, LOS linear, R_a cubic.

Context: Tapping & Morgan (2017); Clette (2021); Okoh & Okoro (2020).

Method: Beta Diagnostic

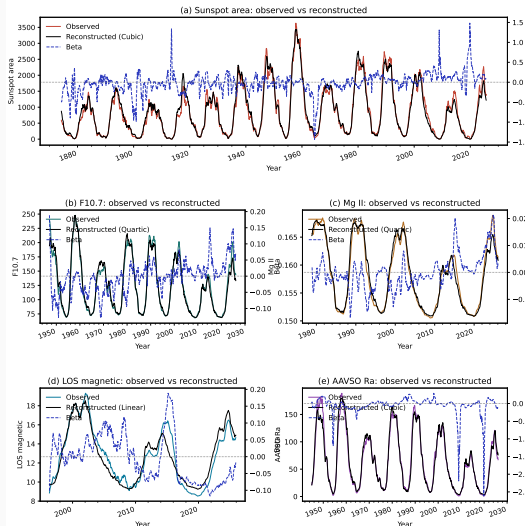
β measures the fractional proxy residual after reconstructing the proxy from SN.

$$\beta(t) = \frac{X_{\text{obs}}(t) - X_{\text{recon}}(t)}{X_{\text{obs}}(t)}$$

- X_{obs} is the measured proxy.
- X_{recon} is the proxy expected from the fitted SN–proxy relation.
- β near zero means the proxy behaves as expected from SN.
- Sustained non-zero β means the relation is unusual.

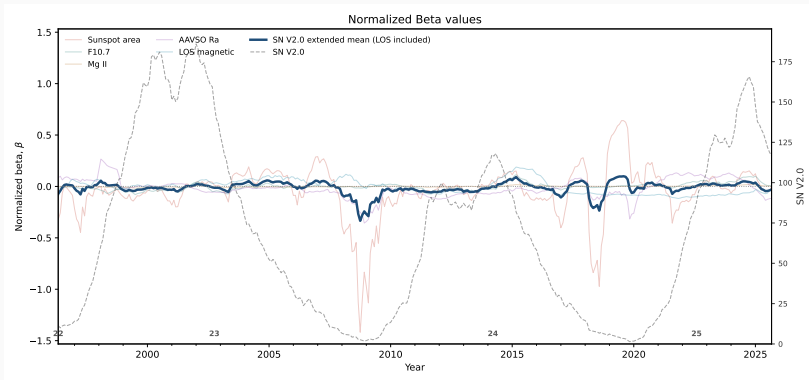
This follows the broad literature practice of comparing departures from expected SN–proxy relations; the exact β definition is the diagnostic used in this workflow.

Beta Shows When Relations Change



- Month-scale excursions are expected.
- Sustained structure is more important than isolated noise.
- Same-sign behavior across proxies is the signal to follow.

Mean Value of Beta Stays Small on Overlapping Periods



- Small value of mean
=> overall SN-proxy
relation is stable.

Steps for Detecting Drift: Detection

1 Data

Align monthly data

Common timestamps, monthly means, smoothing (centered 10 months running mean).

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Deviation-Detection

1. **Use monthly SN and proxy series, with the selected SN→proxy fit for each proxy.**

2. **Compute residual diagnostic**

For each proxy, use the fractional residual:

$$\beta(t) = \frac{X_{\text{obs}}(t) - X_{\text{recon}}(t)}{X_{\text{obs}}(t)}.$$

3. **Put all proxies on a common scale**

Standardize each proxy residual using median/MAD scaling, then apply a centered 6-month running mean.

4. **Test ensemble coherence**

Flag months where the deviation is strong, persistent, and shared by several proxies with the same sign.

Detection rule

A coherent interval requires:

- median deviation strength ≥ 1.5
- at least 3 coherent proxies
- run length ≥ 4 months

Note

This gives where the usual SN–proxy relation becomes unusually organized across proxies.

Deviation Detection

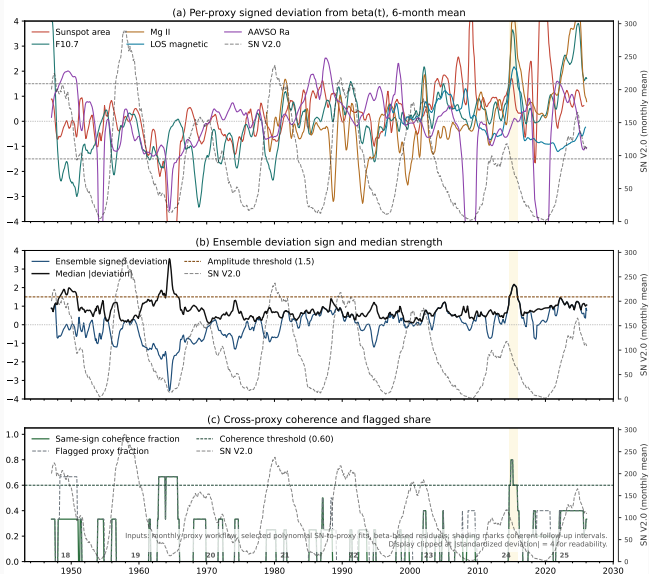


Figure panels

- **Panel (a):** per-proxy signed deviations
- **Panel (b):** ensemble sign and strength
- **Panel (c):** cross-proxy coherence

Yellow shading marks coherent follow-up intervals.

- Coherent interval: 2014-09 to 2015-11.
- Proxies: F10.7, LOS magnetic, Mg II, and Sunspot area.

Does SN V2.0 show coherent drift relative to independent solar-activity proxies?

- Yes. But,
- Deviation are confined to one prominent coherent interval.
- Interval: 2014-09 to 2015-11, Declining phase of the solar cycle 24.
- Proxies: F10.7, LOS magnetic, Mg II, and Sunspot area.

Next Question

Are these deviations solar-origin?

Are These Deviations Solar-Origin?

1. Literature: SN–F10.7

- **Calibration view:** Clette argues for an F10.7 inhomogeneity near 1980–1981 [Clette2021].
- **Solar-physics view:** Mursula et al. argue that changing SN–F10.7 relations reflect real changes in the solar atmosphere, not simply an F10.7 calibration problem [Mursula2024].
- Svalgaard frames the 2014–2015 SN–F10.7 anomaly as a possible SN problem [Svalgaard2025].

2. Literature: SN–area

- Tapping and Morgan show that the relation between SN, total sunspot area, and F10.7 changed noticeably in Cycles 21–24 [TappingMorgan2017].
- A changing sunspot-size distribution can naturally change the SN–area mapping
- Deficits of small spots and multiple sunspot populations support a cycle-dependent SN–area relation.

[LefevreClette2011; Nagovitsyn2016; MunozJaramillo2015; NagovitsynPevtsov2021]

Note

- The literature allows both readings: instrumental/index issues *and* real solar changes.
- Increasingly strong evidence towards a solar-origin deviation.
- Our follow-up tests which interpretation is more consistent with the 2014–2015 deviation window.

Are These Deviations Solar-Origin? Follow-up

- **1. Are high proxy/SN values accompanied by real photospheric or magnetic structures?**
or
Whether statistical departures/deviations correspond to real solar structures?
- **2. Are the 2014–2015 areas high relative to comparable phases of other solar cycles?**

Follow-up purpose

Use independent image and areas to decide whether the deviations are plausibly solar in origin.

Are These Deviations Solar-Origin? Check High Proxy/SN Days

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Use independent image and areas to decide whether the deviations are plausibly solar in origin.

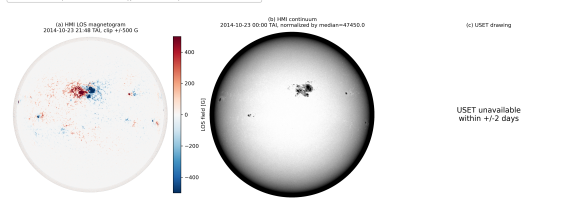
Are These Deviations Solar-Origin? Check High Proxy/SN Days

1. Daily image inspections

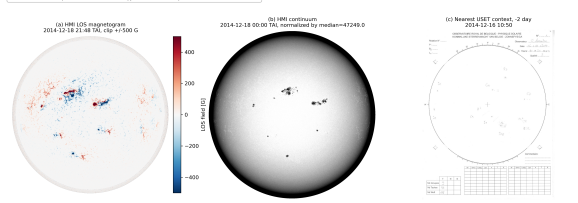
- Candidate windows were selected using high $\text{area}/(\text{SN}+1)$, $|\text{HMI } B|/(\text{SN}+1)$, or joint score.
- For each selected date, we inspected HMI LOS magnetograms, HMI continuum images, and USET drawings where available.
- Several windows show large sunspots, active regions, or extended magnetic-field patches.

High Area per Sunspot Number

Best date: 2014-10-23 | SN: 132.0 | Corrected area: 5364.5 MSH | HMI unsigned LOS: 2.75×10^8
Area/(SN+1): 40.33 | HMI/(SN+1): 2.07×10^6 | Joint score: 40.26 | Window rank: 188



Best date: 2014-12-18 | SN: 154.0 | Corrected area: 3388.5 MSH | HMI unsigned LOS: 2.68×10^8
Area/(SN+1): 21.86 | HMI/(SN+1): 1.73×10^6 | Joint score: 30.74 | Window rank: 310



2014-10-23

- SN = 132
- Corrected area = 5364.5 MSH
- Area/(SN+1) = 40.33
- HMI unsigned LOS = 2.75×10^8

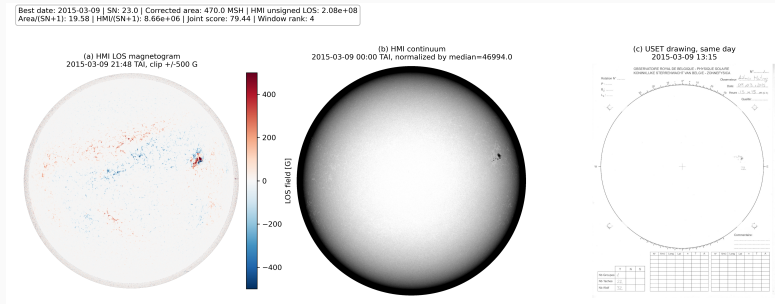
2014-12-18

- SN = 154
- Corrected area = 3388.5 MSH
- Area/(SN+1) = 21.86
- HMI unsigned LOS = 2.68×10^8

Note

High area/SN windows coincide with large continuum structures and strong magnetic-field patches.

Low SN but Clear Magnetic Structure



2015-03-09

- SN = 23
- Corrected area = 470.0 MSH
- $\text{Area}/(\text{SN}+1) = 19.58$
- $\text{HMI}/(\text{SN}+1) = 8.66 \times 10^6$

Note

The proxy excess is physically traceable to solar magnetic and white-light structure.

Even with modest SN, the magnetogram and continuum show a compact active region, also visible in the same-day USET drawing.

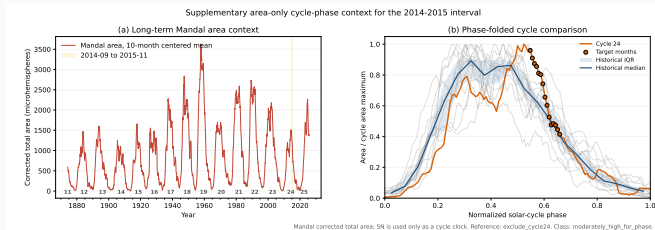
Are These Deviations Solar-Origin? Is the 2014–2015 Period Unusual?

- 1. Are high proxy/SN values accompanied by real photospheric or magnetic structures?
or
Whether statistical departures/deviations correspond to real solar structures?
- 2. Are the 2014–2015 areas high relative to comparable phases of other solar cycles?

Follow-up purpose

Use independent image and areas to decide whether the deviations are plausibly solar in origin.

Is the 2014–2015 Period Unusual?



What this test does

- Compares Cycle 24 area evolution against the historical phase-folded area distribution.
- Tests whether the 2014–2015 deviation months are high for their solar-cycle phase.

Note

- The deviation window is not only a proxy/SN residual feature.
- It also corresponds to elevated sunspot area relative to comparable cycle phases.

The 2014–2015 period shows areas are **moderately high for phase**.

Are These Deviations Solar-Origin? Summary

- **1. Are high proxy/SN values accompanied by real photospheric or magnetic structures?**
or
Whether statistical departures/deviations correspond to real solar structures?
 - The proxy excess is physically traceable to solar magnetic and white-light structure.
- **2. Are the 2014–2015 areas high relative to comparable phases of other solar cycles?**
 - Deviation period corresponds to elevated sunspot area relative to comparable solar-cycle phases.

Does SN V2.0 show coherent drift relative to independent solar-activity proxies?

- Yes. But, Deviation are confined to one prominent coherent interval.

Are these deviations solar-origin?

- Most probably, combined evidence and literature strongly suggests solar-origin.

Questions?

References: Sunspot Number and Proxy Context

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